SDC 3: Characteristics of Included Articles on Artificial Intelligence in Plastic Surgery

3.1 Aesthetic and Breast Surgery

Author, Year	Journal	Clinical Application	Study Aim	Summary of Findings	Study Type	Study Phase	Limitations						
							Descriptive and Data	*Statistical Performance*	*Observational Workflow Analysis*	*Technical Performance and Safety*	*Performance in a Medical Setting*	*Effectiveness in Cohort*	*Total Limitations*
Xie, 2007[27]	Computer Methods and Programs in Biomedicine	Pre-op Consultation	Surgical Planning of Rhinoplasty	Unsupervised ML to simulate rhinoplasty preoperatively and detect all regions of collisions between facial bones and the implant model. Model was able to identify all actively colliding regions; stable over time.	Algorithm development and performance evaluation	0	0.12	0.2, 0.21, 0.22					4
Talbar, 2019[35]	Bio-Algorithms and Med-Systems	Post-op Outcomes	Facial Recognition Post Operatively	Use AGLOH-based feature extraction and ANN classification to validate identities of 452 subjects after plastic surgery based on preoperative and postoperative photographs. Accuracy of feature extraction ranged between plastic surgery procedures, but overall was extremely high.	Algorithm development and performance evaluation	0		0.2		1.1			2
Gunes, 2006[70]	International Journal of Human-Computer Studies	Pre-op Consultation	Objective Evaluation of Aesthetics of Facial Plastic Surgery	Supervised ML to automatically and objectively grade facial beauty based on automatically extracted facial features from diverse female facial images. ML accurately classified beauty compared to "ground truth" of 48 diverse referees.	In silico algorithm performance optimization	1				1.4	2.5		2
Silva, 2019[30]	Institute of Electrical and Electronics Engineers	Post-op Outcomes	Objective Evaluation of Aesthetics of Breast Reconstruction	DNN Automatic detection of breast key points from photographs of 221 breast reconstructions to evaluate aesthetic outcomes. DNN was superior to hybrid model in detection of all breast key points aside from contour.	In silico algorithm performance optimization	1				1.2	2, 2.1, 2.5		4
Borsting, 2020[31]	The Journal of Craniofacial Surgery	Post-op Outcomes	Objective Evaluation of Aesthetics of Rhinoplasty	DCNN to classify images as pre or post rhinoplasty status. DCNN predicted rhinoplasty status was not statistically significantly different from predictions of plastic surgery residents and fellows.	In silico algorithm performance optimization	1				1, 1.1	2.5		3
Geras, 2017[28]	arXiv	Pre-op Consultation	Detection of Breast Cancer Risk	Multi-view Deep Convolutional NN (MV-DCN) to predict BI-RADS level (and probability of prediction) based on mammography, tested on the scans of 129,208 patients. MV-DCN had comparable predictions to four human expert radiologists, accuracy of prediction increased using combination of MV-DCN and radiologists.	Algorithm performance in a medical setting	2				1	2, 2.1		3
Guan, 2019[29]	Journal of Medical Imaging	Pre-op Consultation	Improve AI for Detection of Breast Cancer Risk	GAN used to improve CNN accuracy in breast cancer detection in screening via mammography by generating synthetic labeled lesions for CNN training. GAN generated images were superior to augmented images. Synthetic data can improve reliability of predictive CNN in rare conditions with less available data.	Quality Improvement	2		0.22			2, 2.2		4
Patcas, 2019[32]	International Journal of Oral & Maxillofacial Surgery	Post-op Outcomes	Objective Evaluation of Aesthetics and Perceived Age in Facial Plastic Surgery	CNN used to predict age and evaluate facial attractiveness using post-treatment photographs of 146 consecutive orthognathic patients. CNN previously validated and fine-tuned for study was able to detect orthognathic treatment resulted in younger (nearly 1 year) and more attractive appearing faces (74.7%), these differences were especially seen in lower jaw surgery.	Algorithm performance in a medical setting	2					2.1, 2.4		2
O'Neill, 2020[25]	Annals of Surgical Oncology	Predictive analytics	Autologous Breast Reconstruction Clinical Outcomes	Unsupervised ML classification trees to predict flap failure based off clinical characteristics in a retrospective cohort of 1012 patients undergoing microvascular breast surgery (DIEP flap) by classifying high risk patients and identifying contributing risk factors. Decision-tree model identified 10% of patients as belonging to one of four high risk groups; two thirds of flap failures occurred in these high-risk patients. ML identified that mean BMI significantly predicted flap failure, particularly predicting post-operative complications when combined with a subset of other comorbidities.	Algorithm performance in a medical setting	2	0.11, 0.12	0.22			2, 2.4		5
Dorfman, 2020[33]	Aesthetic Surgery Journal	Predictive analytics	Objective Evaluation of Aesthetics and Perceived Age in Facial Plastic Surgery	CNN photographical analysis to detect differences in estimated human age recognition before and after rhinoplasty in 100 patients. CNN accurately and precisely estimated human age (r=0.91) and identified patients looked statistically significantly younger following rhinoplasty (post-operative age decreased by 3 years).	Algorithm performance in a medical setting	2		0.22		1.1	2, 2.2		4
Chen, 2020[34]	Plastic and Reconstructive Surgery	Post-op Outcomes	Objective Evaluation of Aesthetics and Perceived Gender in Facial Feminization Surgery	CNN to confirm perception of female gender from photographs following post-operative feminizing surgery. Accuracy of CNN in 120)= cisgender male and female controls was 100% and 98% respectively. CNN gendered 20 facial	Algorithm performance in a medical setting	2					2, 2.2		2

				feminization patients as female 53% of the time preoperatively and 98% of the time postoperatively, revealing an overall significant improvement of 45%.								
Mavioso, 2020[26]	The Breast	Pre-op Consultation	Preoperative Mapping of Perforators for Breast Reconstruction	Computer Vision of CT angiography accurately mapped perforator branches for 40 patients undergoing DIEP flap reconstruction. Semi-automatic technique significantly reduced surgical time and mean error compared that seen with standard manual mapping.	Clinical Trial	3				1.2, 1.5	3	3
3.2 Craniofacial Surgery												
Xi, 2010[80]	The International Journal of Precision Engineering and Manufacturing	Post-op Outcomes	Objective Evaluation of Cleft Lip Reconstruction	ANN analyzed nostril symmetry from images of 85 patients following cleft lip reconstruction. ANN accuracy increases with increases in training sample set size.	Algorithm development and performance evaluation	0		0.2		1		2
Mendoza, 2014[40]	Medical Image Analysis	Predictive analytics	Diagnosis of Craniosynostosis	Supervised ML (regularized linear discriminant analysis) differentiates types of non-syndromic craniosynostosis based on CT imaging with sensitivity 92.3% and specificity 98.9% accuracy comparable to trained radiologists.	Algorithm development and performance evaluation	0	0.12	0.2		1		3
Wu, 2016[36]	The Cleft Palate-Craniofacial Journal	Pre-op Consultation	Surgical Planning of Cleft Lip Repair	Supervised ML to determine a reference plane and production of 3D image to guide cleft lip reconstruction. Manual methods outperformed supervised ML.	Algorithm development and performance evaluation	0		0.2				1
Ferry, 2019[39]	Elife	Predictive analytics	Diagnosis of Facial Genetic Disorders	Unsupervised ML and computer vision for facial recognition and diagnosis of 8 genetic disorders to aid clinical narrowing of diagnosis by 27.6-fold.	Prototype design	0	0.1, 012					2
Nishimoto , 2019[42]	The Journal of Craniofacial Surgery	Predictive analytics	Preoperative Anatomic Landmarking of Craniomaxillofacial Skeletal Profile	CNN to automatically predict skeletal landmarks on cephalogram images for evaluation of craniomaxillofacial skeletal profile. CNN predicted landmarks were not significantly different from manual landmarks.	Algorithm development and performance evaluation	0	0.1, 0.12, 0.13					3
Bhalodia, 2020[38]	The Journal of Craniofacial Surgery	Pre-op Consultation	Diagnosis of Craniosynostosis	Supervised ML to predict the severity of metopic craniosynostosis. Supervised ML was used to combine a validated shape 3D modeling using 82 CT images and opinions from 18 experts. The ML model effectiveness was statistically significant, and the shape descriptor was accurately predictive of overall severity.	Algorithm development and performance evaluation	0	0.11, 0.12, 0.13		0.31	1		5
Ma, 2020[43]	The International Journal of Medical Robotics and Computer Assisted Surgery	Pre-op Consultation	Preoperative Anatomic Landmarking of Craniomaxillofacial Skeletal Profile	Deep Neural Network + CNN for automatic 3D landmarking of CT images for cephalometric analysis with consideration of anatomical relationships. Automatic 3D landmarking CT images was fulfilled with accuracy of 5.785 mm.	Algorithm development and performance evaluation	0	0.11, 0.12, 0.13	0.2	0.31	1		6
Maier, 2008[78]	The Journal of Acoustical Society of America	Pre-op Consultation	Assess Speech Quality	Supervised ML assesses speech impediments in cleft lip patients in relation to five speech disorders associated with cleft lip and palate. ML was reliable and in the range of human raters.	In silico algorithm performance optimization	1		0.2		1, 1.5		3
Sari, 2008[37]	The Cleft Palate-Craniofacial Journal	Medical Research	Assess Speech Quality	Used previously validated NLP in a case report to detect changes in speech production following cleft lip repair.	Usability test	1		0.22		1.1		2
Dusseldorp, 2019[41]	Plastic and Reconstructive Surgery	Post-op Outcomes	Objective Evaluation of Facial Palsy Reconstruction	Computer vision and ML used to detect perceived emotion in images of patients pre- and postoperative facial palsy reconstruction. AI pre- and post-operative comparative image analysis perceived 84% increase in joy in flaccid facial palsy patients and a significant reduction in negative emotion in non-flaccid palsy patients.	Algorithm performance in a medical setting	2					2, 2.1	2
3.3 Non-breast Microsurgery												
Fichera, 2013[48]	Institute of Electrical and Electronics Engineers	Streamline clinic & Procedures	Intraoperative Supervising System in Laser Microsurgery	Uses image processing and artificial intelligence to automatically supervise laser phono microsurgery. Cognitive Supervisory System is trained with ML of temperature dynamics and cognitive mapping to predict unintentional thermal tissue damage, control incision depth, and generate appropriate alarms in laser phono microsurgery. Preliminary results on chicken tissue samples are positive.	Prototype design	0	0.11	0.22, 0.21		1.1		4
Tatinati, 2015[47]	Institute of Electrical and	Streamline clinic & Procedures	Tremor Filtering of Handheld Surgical Tools	Unsupervised ML (kernel learning) to accurately filter tremulous motion to compensate with robotic handheld surgical	Prototype design	0		0.2				1

	Electronics Engineers			assists in surgery. Machine learning in hand-held instruments outperforms other methods in preliminary studies.							
Power, 2018[49]	Small Journal	Streamline clinic & Procedures	Microstructure Manipulation for Robotic Device	Used ANN to allow for force estimation of 3D force sensor in microscale robotic 3D compliant grasper with the ability to isolate and manipulate microstructures as small as alveoli, villi, or individual cells.	Prototype design	0		0.2	1.5		2
Nakazawa, 2020[45]	International Journal of Computer Assisted Radiology and Surgery	Streamline clinic & Procedures	Intraoperative Detection of Suture Needle Motion	DNN to perform real-time video-based needle detection during microvascular anastomosis. Preliminary results on artificial blood vessels show precision of 89.2%.	Prototype design	0	0.12		1, 1.2, 1.5		4
Kiranantawa, 2014[46]	Plastic and Reconstructive Surgery	Post-op Outcomes	Objective Evaluation of Post-Operative Free Flap Perfusion	SilpaRamanitor used supervised ML (k-nearest neighbor) to categorize perfusion based on smartphone photos of patients.	In silico algorithm performance optimization	1			1, 1.1		2
Kuo, 2018[44]	Oncotarget	Post-op Outcomes	Prediction of Free Flap SSI	(ANN) to predict surgical site infections in microvascular free-flap reconstruction based on pre-op and post-op risk factors in head and neck cancer resection patients. ANN pre-op and post-op predictions significantly outperformed logistic regression.	Algorithm performance in a medical setting	2			1	2, 2.2	3
3.4 Burn Surgery											
Estahbanati, 2002[50]	Burns Journal	Predictive analytics	Prediction of Burn Survival	ANN trained and validated on a retrospective cohort of 1082 acute burn clinical patient data achieved 90% accuracy in predicting outcomes of burn patients.	Algorithm development and performance evaluation	0		0.2, 0.21	1		3
Yeong, 2005[52]	Burns Journal	Predictive analytics	Prediction of Burn Wound Healing Time	ANN uses input reflectance spectrometry to predict burn healing time on a prospective cohort of 39 patients. ANN had diagnostic accuracy of 86%. Direct visualization has an accuracy of 50-70% and laser Doppler flowmetry is 90% accurate.	Algorithm development and performance evaluation	0	0.1, 0.11, 0.12, 0.13	0.2, 0.22			6
Heredia-Juesas, 2016[54]	Institute of Electrical and Electronics Engineers	Streamline clinic & Procedures	Classification of Burn Wound Depth	Supervised ML (quadratic discriminant analysis) to classify burn wound depth for debridement purposes during burn surgery. Accuracy of this system was highest for healthy skin and lowest for shallow burns on animal models.	Algorithm development and performance evaluation	0	0.11	0.2, 0.22	1	2.5	5
Yamamura, 2007[51]	Biomedicine & Pharmacotherapy	Predictive analytics	Prediction of Burn Patient Response to Aminoglycosides	ANN to predict response to aminoglycosides using lab and burn characteristics of 25 patients with burns. Validation on prospective cohort showed ANN correctly predicted the response to ABK 100%, superior to standard (PK/PD) and logistic regression 72% accuracy.	In silico algorithm performance optimization	1	0.12		1.1	2.5	3
Martínez-Jiménez, 2018[53]	PLOS ONE	Predictive analytics	Prediction of Necessary Burn Treatment Modality	Supervised ML to predict appropriate treatment of burns based on initial thermal imaging. Validity study on prospective cohort with blinded surgical teams shows accuracy of 90%.	In silico algorithm performance optimization	1			1.1, 1.3		2
3.5 General Plastic and Reconstructive Surgery											
Sarker, 2015[57]	Artificial Intelligence in Medicine	Medical Research	Enhance and Improve Efficiency in Research	Natural language processing to find relevant information within medical research to synthesize quality evidence.	In silico algorithm performance optimization	1			1, 1.1, 1.2		3
Baur, 2018[23]	arXiv	Pre-op Consultation	Improve AI for Detection of Dermatologic Malignancy	Generative adversarial network (PGAN) to produce malignant dermatoscopic images. Validation study showed dermatologist and deep learning experts were not able to significantly discern real images from fake images.	In silico algorithm performance optimization	1			1.2	2.1	2
Levites, 2019[55]	Plastic and Reconstructive Surgery	Streamline clinic & Procedures	Analyze Social Media Trends of Emotional Engagement with Plastic Surgery	NLP and supervised ML to detect and quantify emotional engagement with specific plastic surgery terms via Twitter for insight into plastic surgery patient motivations.	Usability test	1		0.22	1.1	2.4	3
Boczar, 2020[56]	Annals of Plastic Surgery	Streamline clinic & Procedures	Virtual Assistant for Plastic Surgery	NLP to train a virtual assistant software to answer frequently asked questions pertaining to plastic surgery. Participants of the study considered the software's accuracy to be 92.3%, and correct 83.3% of the time	Usability test	1			1.1		1
3.6 Oral & Maxillofacial Surgery											
Knoops, 2019[60]	Scientific Reports	Predictive analytics	Predict Postoperative Orthognathic Shape Features	Used a combination of ML computer vision and supervised learning to create clinical 3D morphable models that classified and diagnose patients with orthognathic shape features as indications for orthognathic surgery as well as simulate post-operative face shapes.	Algorithm development and performance evaluation	0	0.11, 0.12, 0.13		1.4	2.5	5
Zhang, 2018[58]	Scientific Reports	Post-op Outcomes	Prediction of Postoperative Facial Swelling	Use of ANN and patient oral/ facial parameters to predict severity of postoperative facial swelling following impacted mandibular third molar extraction with 98.00% predictive accuracy. n=400 patients: 300 training and 100 test sets.	Algorithm performance in a medical setting	2		0.2		2, 2.5	3

Stehrer, 2019[59]	Journal of Cranio-Maxillo-Facial Surgery	Pre-op Consultation	Prediction of Blood Loss in Orthognathic Surgery	Supervised ML (random forest) to predict exact mL of perioperative blood loss in orthognathic surgery based on patient characteristics. Predictive model had statistically significant accuracy and revealed that bimaxillary surgery and preoperative hematocrit had the greatest feature importance. n=950 patients (80% training set and 20% testing set)	Algorithm performance in a medical setting	2					2	1
3.7 Wound Healing													
Robnik-Sikonja, 2003[79]	Artificial Intelligence in Medicine	Predictive analytics	Prediction of Wound Healing Time	Supervised ML to rank prognostic factors and accurately predict wound healing time in 214 patients, (300 wounds) after 4 weeks. Study found most significant prognostic factor was initial wound area. Predictive accuracy increased directly with follow-up time.	Algorithm performance in a medical setting	2					2	1
Jung, 2016[61]	Wound Repair and Regeneration	Predictive analytics	Prediction of Chronic Wound Healing Time	Supervised ML to identify prognostic factors for and accurate prediction of delayed chronic wound healing. ML trained on EHR data from 53,354 patients, with a total of 150,277 wounds across 68 wound care centers. Found concurrent palliative care was the most important predictor. Agreement in predictive probabilities were significant.	Algorithm performance in a medical setting	2				1.1	2, 2.3, 2.4	4
3.8 Hand Surgery													
Luján, 2004[64]	Medical & Biological Engineering & Computing	Streamline clinic & Procedures	Stimulation of Muscle Movement for Control of Hand and Wrist Movement in Neuroprosthesis	ANN was used to develop neuroprosthesis with improved hand and wrist mobility for patients with spinal cord injury. Preliminary study on a single subject showed limited success, with better predictive abilities than hand control.	Prototype design	0		0.2, 0.21, 0.22	0.3			4
Hincapie, 2009[68]	Institute of Electrical and Electronics Engineers	Streamline clinic & Procedures	Prediction of Muscle Activation for the Development of Neuroprosthesis	ANN trained to predict voluntary muscle activations for development of neuroprosthesis controller in spinal cord injury.	Algorithm development and performance evaluation	0	0.11, 0.12, 0.13	0.22	0.3	1.2, 1.4, 1.5		8

Table 2: Characteristics of Included Articles on Artificial Intelligence in Plastic Surgery. This table qualitatively describes the included articles in this study is grouped by plastic surgery subspecialty and phase of artificial intelligence in health care, sorted chronologically by year. This order is kept consistent throughout the tables and figures. The study types as described by Park et al. and modified for this paper include algorithm development and performance evaluation and prototype design (phase 0), prototype design (phase 0), in silico algorithm performance optimization (phase 1), usability test (phase 1), interface design (phase 2), quality improvement (phase 2), clinical trial (phase 3), adverse events identification (phase 3), post deployment surveillance (phase 4). These study types along with the previously defined limitations for AI in healthcare, were used to consistently describe limitations of each study through qualitative analysis for the purposes of post-hoc analysis. The limitations assigned to each study were categorized by previously described obstacles for various phases of AI in healthcare and adapted for this review. The values were coded according to the following categories: Descriptive and Data Limitations: 0.1 (data quality), 0.11 (data representativeness), 0.12 (data completeness), 0.13 (data validity). Statistical Performance Limitations: 0.2 (statistical performance metrics of algorithm), 0.21 (optimization of model performance (determination of tradeoff between precision and recall)), 0.22 (human comparison). 0.3 (identification of user needs) 0.31 (work-flow issues). Technical Performance and Safety Limitations: 1 (human resource intensive), 1.1 (generalizability), 1.2 (optimizing implementation of AI), 1.3 (determination of useful functionalities and design options), 1.4 (understanding social and cultural background which affects clinical decision making/data timeliness), 1.5 (prototyping initial design). Performance in a Medical Setting Limitations: 2 (validating improvement of intermediate measures to justify larger trials), 2.1 (impact of artifacts from simulated scenarios), 2.2 (require controlled experiment provide effect size estimate), 2.3 (full context of clinical workflow and mental models), 2.4 (cannot estimate effect in real-world scenarios trusting the AI algorithm), 2.5 (testing with cross-validation design). Effectiveness in Cohort Limitations: 3 (evaluation of effectiveness in prospective cohorts).